# Palmer Penguins Analysis

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## Introduction

This report explores the **Palmer Penguins dataset**, which contains observations of 344 penguins from three species: Adelie, Chinstrap, and Gentoo, across three islands in the Palmer Archipelago, Antarctica. The data were collected by Dr. Kristen Gorman.

The purpose of this analysis is to understand how these species differ in their physical characteristics and how these differences relate to their environment. The focus is placed on body size, flipper dimensions, and bill structure, as well as how these features vary across islands.

The analysis begins with an exploration of the dataset, followed by summary statistics to compare species. Visualizations and statistical tests are then used to uncover relationships between variables. This step by step approach allows patterns in the data to emerge clearly.

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## Setup

```r
library(tidyverse)
library(palmerpenguins)
```

---

## Data Exploration

To begin, the dataset is inspected to understand its structure and contents.

```r
head(penguins)
```

```
## # A tibble: 6 x 8
##   species island    bill_length_mm bill_depth_mm flipper_length_mm body_mass_g
##   <fct>   <fct>              <dbl>         <dbl>             <int>       <int>
## 1 Adelie  Torgersen           39.1          18.7               181        3750
## 2 Adelie  Torgersen           39.5          17.4               186        3800
## 3 Adelie  Torgersen           40.3          18                 195        3250
## 4 Adelie  Torgersen           NA            NA                  NA          NA
## 5 Adelie  Torgersen           36.7          19.3               193        3450
## 6 Adelie  Torgersen           39.3          20.6               190        3650
## # i 2 more variables: sex <fct>, year <int>
```

```r
nrow(penguins)
```

```
## [1] 344
```

```r
ncol(penguins)
```

```
## [1] 8
```

```r
glimpse(penguins)
```

```
## Rows: 344
## Columns: 8
## $ species           <fct> Adelie, Adelie, Adelie, Adelie, Adelie, Adelie, Adel~
## $ island            <fct> Torgersen, Torgersen, Torgersen, Torgersen, Torgerse~
## $ bill_length_mm    <dbl> 39.1, 39.5, 40.3, NA, 36.7, 39.3, 38.9, 39.2, 34.1, ~
## $ bill_depth_mm     <dbl> 18.7, 17.4, 18.0, NA, 19.3, 20.6, 17.8, 19.6, 18.1, ~
## $ flipper_length_mm <int> 181, 186, 195, NA, 193, 190, 181, 195, 193, 190, 186~
## $ body_mass_g       <int> 3750, 3800, 3250, NA, 3450, 3650, 3625, 4675, 3475, ~
## $ sex               <fct> male, female, female, NA, female, male, female, male~
## $ year              <int> 2007, 2007, 2007, 2007, 2007, 2007, 2007, 2007, 2007~
```

The dataset contains 344 observations and 8 variables, including both numerical measurements and categorical groupings such as species and island.

### Checking for Missing Values

Before proceeding, missing values are examined.

```r
colSums(is.na(penguins))
```

```
##           species            island    bill_length_mm     bill_depth_mm
##                 0                 0                 2                 2
## flipper_length_mm       body_mass_g               sex              year
##                 2                 2                11                 0
```

Missing values are relatively few and occur mainly in measurement variables and sex. These are handled using `na.rm = TRUE` or filtering where necessary to ensure accurate calculations.

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## Calculations

### Summary Statistics by Species

To compare the physical characteristics of each species, summary statistics are calculated.

```r
penguins %>%
  group_by(species) %>%
  summarise(
    avg_bill_length = mean(bill_length_mm, na.rm = TRUE),
    mid_bill_length = median(bill_length_mm, na.rm = TRUE),
    min_bill_length = min(bill_length_mm, na.rm = TRUE),
    max_bill_length = max(bill_length_mm, na.rm = TRUE),
    sd_bill_length = sd(bill_length_mm, na.rm = TRUE),
    avg_flipper_length = mean(flipper_length_mm, na.rm = TRUE),
    mid_flipper_length = median(flipper_length_mm, na.rm = TRUE),
    min_flipper_length = min(flipper_length_mm, na.rm = TRUE),
    max_flipper_length = max(flipper_length_mm, na.rm = TRUE),
    sd_flipper_length = sd(flipper_length_mm, na.rm = TRUE)
  )
```

```
## # A tibble: 3 x 11
##   species   avg_bill_length mid_bill_length min_bill_length max_bill_length
##   <fct>               <dbl>           <dbl>           <dbl>           <dbl>
## 1 Adelie               38.8            38.8            32.1            46
## 2 Chinstrap            48.8            49.6            40.9            58
## 3 Gentoo               47.5            47.3            40.9            59.6
## # i 6 more variables: sd_bill_length <dbl>, avg_flipper_length <dbl>,
## #   mid_flipper_length <dbl>, min_flipper_length <int>,
## #   max_flipper_length <int>, sd_flipper_length <dbl>
```

The results show clear differences across species. Gentoo penguins generally have larger measurements, while Adelie penguins are smaller. This indicates that species plays a key role in determining physical traits.

---

### Average Body Mass and Flipper Length by Species

To further examine size differences, body mass and flipper length are compared.

```r
penguins %>%
  group_by(species) %>%
  summarise(
    avg_body_mass = mean(body_mass_g, na.rm = TRUE),
    avg_flipper_length = mean(flipper_length_mm, na.rm = TRUE)
  ) %>%
  arrange(-avg_body_mass)
```

```
## # A tibble: 3 x 3
##   species   avg_body_mass avg_flipper_length
##   <fct>             <dbl>              <dbl>
## 1 Gentoo            5076.               217.
## 2 Chinstrap         3733.               196.
## 3 Adelie            3701.               190.
```

Gentoo penguins have the highest average body mass and longest flippers, confirming that they are the largest species in the dataset.

---

### Heaviest Island by Average Body Mass

The analysis now considers whether location influences body mass.

```
penguins %>%
  group_by(island) %>%
  summarise(avg_body_mass = mean(body_mass_g, na.rm = TRUE)) %>%
  arrange(-avg_body_mass)
```

```
## # A tibble: 3 x 2
##   island    avg_body_mass
##   <fct>             <dbl>
## 1 Biscoe            4716.
## 2 Dream             3713.
## 3 Torgersen         3706.
```

Biscoe Island has the highest average body mass. This is likely influenced by the presence of Gentoo penguins, suggesting that species distribution affects island-level characteristics.

---

### Bill Ratio

To better capture differences in bill shape, a new variable is created.

```
penguins <- penguins %>%
  mutate(bill_ratio = bill_length_mm / bill_depth_mm)
head(penguins)
```

```
## # A tibble: 6 x 9
##   species island    bill_length_mm bill_depth_mm flipper_length_mm body_mass_g
##   <fct>   <fct>              <dbl>         <dbl>             <int>       <int>
## 1 Adelie  Torgersen           39.1          18.7               181        3750
## 2 Adelie  Torgersen           39.5          17.4               186        3800
## 3 Adelie  Torgersen           40.3          18                 195        3250
## 4 Adelie  Torgersen           NA            NA                  NA          NA
## 5 Adelie  Torgersen           36.7          19.3               193        3450
## 6 Adelie  Torgersen           39.3          20.6               190        3650
## # i 3 more variables: sex <fct>, year <int>, bill_ratio <dbl>
```

```
penguins %>%
  select(species, bill_length_mm, bill_depth_mm, bill_ratio) %>%
  head()
```

```
## # A tibble: 6 x 4
##   species bill_length_mm bill_depth_mm bill_ratio
##   <fct>            <dbl>         <dbl>      <dbl>
## 1 Adelie            39.1          18.7       2.09
## 2 Adelie            39.5          17.4       2.27
## 3 Adelie            40.3          18         2.24
## 4 Adelie            NA            NA        NA
## 5 Adelie            36.7          19.3       1.90
## 6 Adelie            39.3          20.6       1.91
```

This ratio reflects how long and narrow a bill is relative to its depth.

---

### Bill Ratio Analysis

```
penguins %>%
  group_by(species) %>%
  summarise(avg_bill_ratio = mean(bill_ratio, na.rm = TRUE)) %>%
  arrange(-avg_bill_ratio)
```

```
## # A tibble: 3 x 2
##   species   avg_bill_ratio
##   <fct>              <dbl>
## 1 Gentoo              3.18
## 2 Chinstrap           2.65
## 3 Adelie              2.12
```

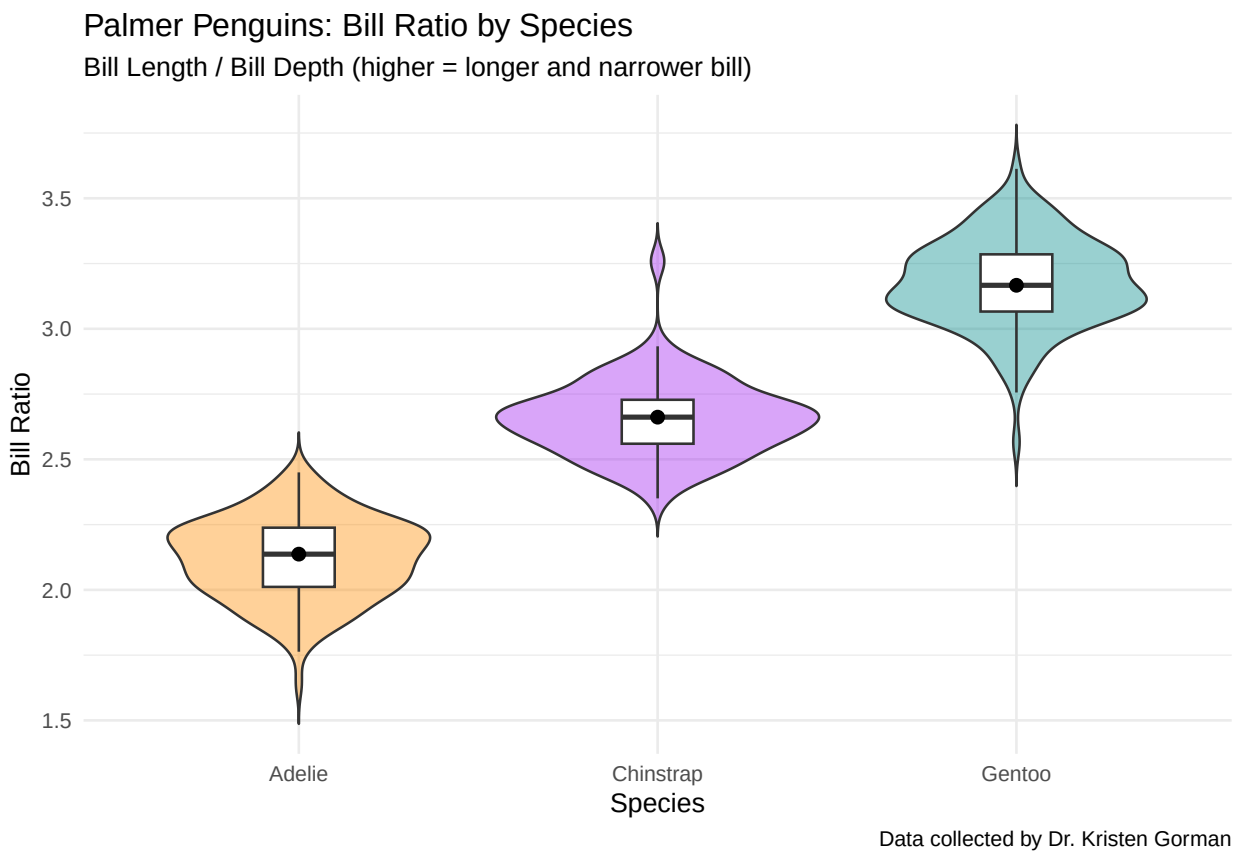
Gentoo penguins have the highest bill ratio, while Adelie penguins have the lowest. This shows clear variation in bill structure across species.

---

## Bill Ratio Visualization

To better understand these differences, the distribution of bill ratios is visualized.

```
ggplot(data = penguins) +
  geom_violin(mapping = aes(x = species, y = bill_ratio, fill = species),
    alpha = 0.4,
    trim = FALSE,
    na.rm = TRUE) +
  geom_boxplot(mapping = aes(x = species, y = bill_ratio),
    width = 0.2,
    fill = "white",
    outlier.shape = NA,
    na.rm = TRUE) +
  stat_summary(mapping = aes(x = species, y = bill_ratio),
    fun = "median",
    geom = "point",
    color = "black",
    size = 2,
    na.rm = TRUE) +
  scale_fill_manual(values = c("darkorange", "purple", "cyan4")) +
  labs(title = "Palmer Penguins: Bill Ratio by Species",
    subtitle = "Bill Length / Bill Depth (higher = longer and narrower bill)",
    x = "Species",
    y = "Bill Ratio",
    caption  = "Data collected by Dr. Kristen Gorman") +
  theme_minimal(base_size = 10) +
  theme(legend.position = "none")
```




The visualization shows distinct distributions for each species, reinforcing that bill structure differs consistently across species.

---

### Male vs Female Count by Species

The distribution of male and female penguins is examined next.

```
penguins %>%
  filter(!is.na(sex)) %>%
  count(species, sex)
```

```
## # A tibble: 6 x 3
##   species   sex        n
##   <fct>     <fct>  <int>
## 1 Adelie    female    73
## 2 Adelie    male      73
## 3 Chinstrap female    34
## 4 Chinstrap male      34
## 5 Gentoo    female    58
## 6 Gentoo    male      61
```

Counts show that males and females are evenly represented across most species.

---

### Male vs Female Percentage by Species

```r
penguins %>%
  filter(!is.na(sex)) %>%
  group_by(species, sex) %>%
  summarise(count = n(), .groups = "drop_last") %>%
  mutate(percentage = count / sum(count) * 100)
```

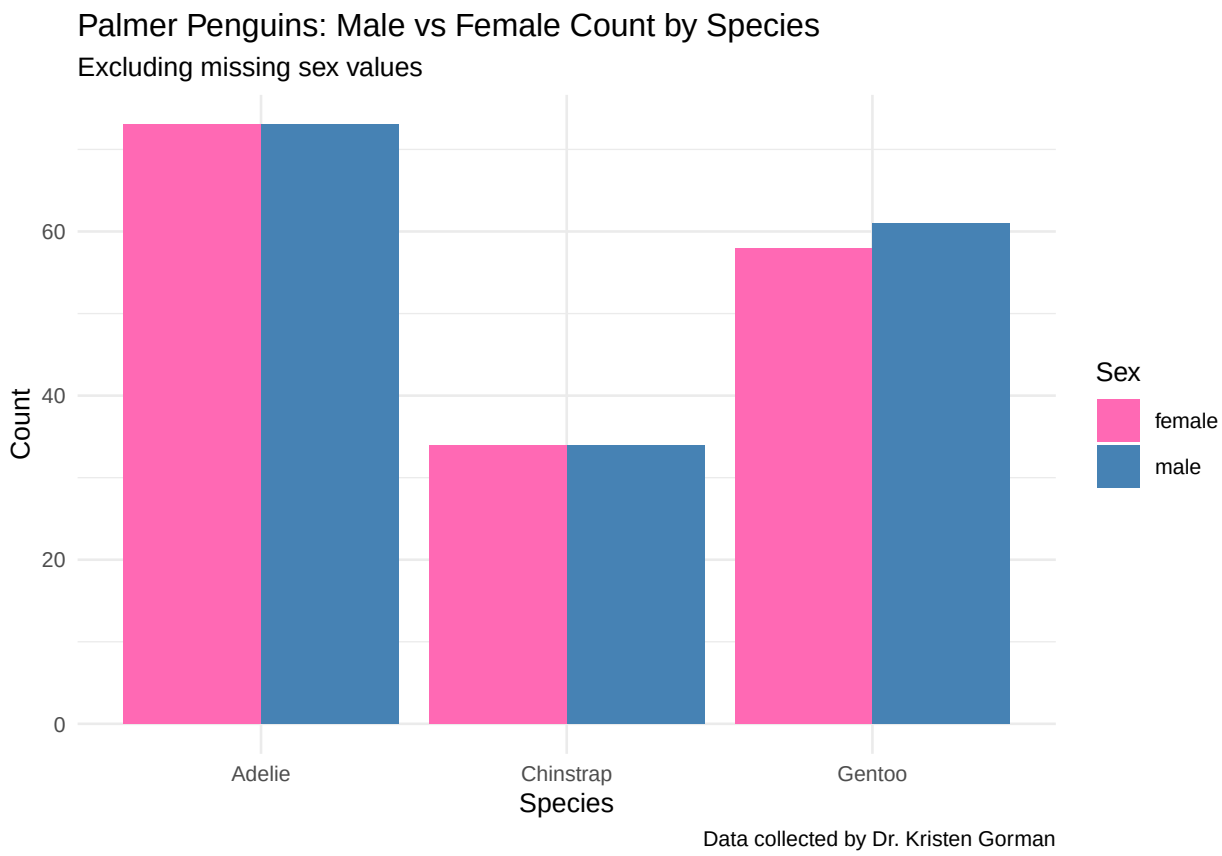
```
## # A tibble: 6 x 4
## # Groups:   species [3]
##   species   sex    count percentage
##   <fct>     <fct>  <int>      <dbl>
## 1 Adelie    female    73       50
## 2 Adelie    male      73       50
## 3 Chinstrap female    34       50
## 4 Chinstrap male      34       50
## 5 Gentoo    female    58       48.7
## 6 Gentoo    male      61       51.3
```

The percentages show an almost perfectly balanced distribution for Adelie and Chinstrap penguins at 50 percent each. Gentoo penguins show a slight difference, with 48.7 percent female and 51.3 percent male. Overall, the distribution remains nearly equal across all species.

---

### Male vs Female Visualization

```r
penguins %>%
  filter(!is.na(sex)) %>%
  ggplot(mapping = aes(x = species, fill = sex)) +
  geom_bar(position = "dodge") +
  labs(title = "Palmer Penguins: Male vs Female Count by Species",
       subtitle = "Excluding missing sex values",
       x = "Species",
       y = "Count",
       fill = "Sex",
       caption = "Data collected by Dr. Kristen Gorman") +
  scale_fill_manual(values = c("female" = "hotpink", "male" = "steelblue")) +
  theme_minimal(base_size = 10)
```




The visualization confirms the balanced distribution observed in the summary statistics.

---

**Island Biodiversity**

Species distribution across islands is examined.

```
penguins %>%
  group_by(island, species) %>%
  summarise(count = n(), .groups = "drop")
```

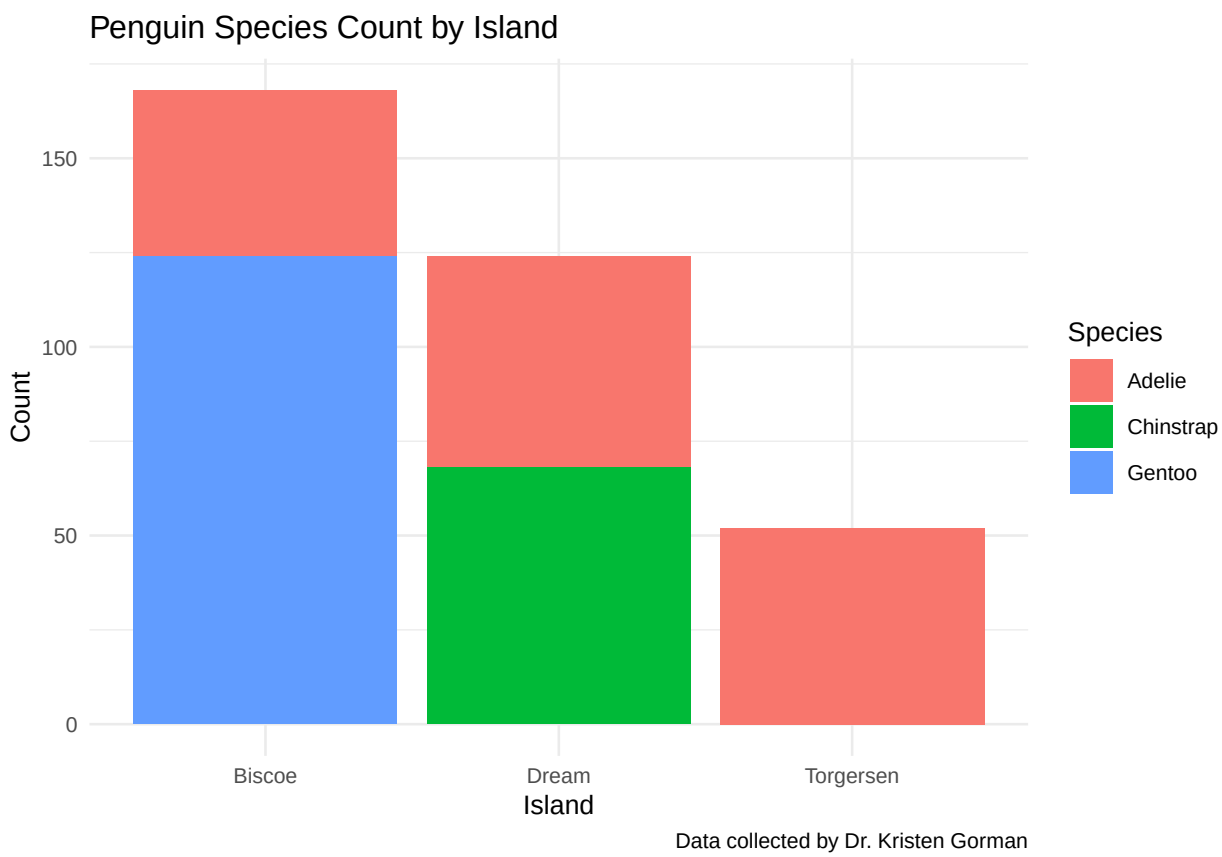
```
## # A tibble: 5 x 3
##   island    species   count
##   <fct>     <fct>     <int>
## 1 Biscoe    Adelie       44
## 2 Biscoe    Gentoo      124
## 3 Dream     Adelie       56
## 4 Dream     Chinstrap    68
## 5 Torgersen Adelie       52
```

This shows how species are spread across locations.

---

**Island Biodiversity Visualization**

```
penguins %>%
  group_by(island, species) %>%
  summarise(count = n(), .groups = "drop") %>%
  ggplot(mapping = aes(x = island, y = count, fill = species)) +
  geom_bar(stat = "identity") +
  labs(title = "Penguin Species Count by Island",
       x = "Island",
       y = "Count",
       fill = "Species",
       caption = "Data collected by Dr. Kristen Gorman") +
  theme_minimal(base_size = 10)
```



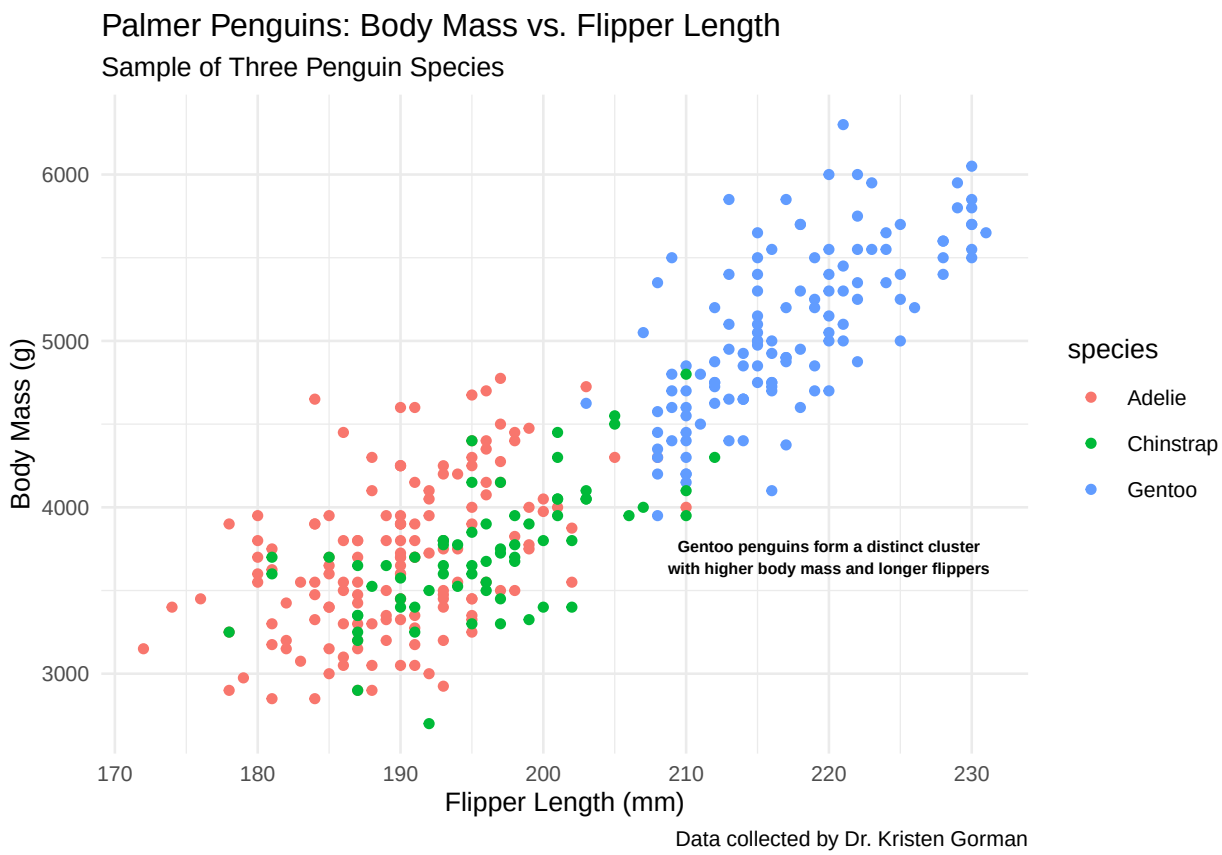

Biscoe and Dream Islands host multiple species, while Torgersen Island contains only Adelie penguins. This indicates differences in biodiversity across islands.

---

## Body Mass vs. Flipper Length Visualization

Earlier results showed that Gentoo penguins tend to have both higher body mass and longer flippers. This raises the question of whether these two variables are related across all species.

```r
ggplot(data = penguins) +
  geom_point(mapping = aes(x = flipper_length_mm, y = body_mass_g, colour = species), na.rm = TRUE) +
  labs(title = "Palmer Penguins: Body Mass vs. Flipper Length",
       x = "Flipper Length (mm)",
       y = "Body Mass (g)",
       subtitle = "Sample of Three Penguin Species",
       caption = "Data collected by Dr. Kristen Gorman") +
  annotate("text", x = 220, y = 3700,
           label =
             "Gentoo penguins form a distinct cluster\nwith higher body mass and longer flippers",
           colour = "black",
           fontface = "bold",
           size = 2) +
  theme_minimal(base_size = 10)
```




The scatter plot shows a clear positive relationship, with larger penguins tending to have longer flippers. Distinct clusters also appear for each species.

### Correlation Test: Flipper Length vs Body Mass

```r
cor.test(penguins$flipper_length_mm, penguins$body_mass_g,
         use = "complete.obs")
```

```
##
```

```
##  Pearson's product-moment correlation
##
## data:  penguins$flipper_length_mm and penguins$body_mass_g
## t = 32.722, df = 340, p-value < 2.2e-16
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  0.843041 0.894599
## sample estimates:
##       cor
## 0.8712018
```

The correlation test confirms a strong positive relationship, indicating that the observed pattern is statistically significant.

---

## Conclusions

The analysis reveals clear differences among penguin species in terms of size, bill structure, and distribution across islands. Gentoo penguins are the largest species, while Adelie penguins are smaller in comparison.

A strong relationship exists between flipper length and body mass, supported by both visualization and statistical testing. Species are also unevenly distributed across islands, contributing to differences in biodiversity.

Overall, the results demonstrate how exploratory data analysis and visualization can uncover meaningful patterns in biological data.

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## What I Learned

- How to perform exploratory data analysis using tidyverse in R
- How to handle missing values effectively
- How to create and interpret multiple types of visualizations
- How to engineer new variables using mutate
- How to quantify relationships using correlation tests
- The importance of structured and step by step analysis

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*Analysis completed by Daniel Eniola | Data source: palmerpenguins R package*